

AHMED GHARIB

Principal Data & AI Engineer

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Professional Summary

Data Engineering and AI professional with 11+ years building scalable data platforms, high-performance pipelines, and intelligent automation across Azure, Snowflake, AWS, GCP, and Databricks. Specialist in Medallion architecture (DBT), async ETL engineering (68x performance gains), and Agentic AI systems (Azure OpenAI GPT, LangChain, N8N, Claude Code). Author of an open-source Dynamics 365 automation CLI designed as an AI-agent surface, and an AIWorx instructor training 150+ developers to build with AI coding agents. Track record of measurable impact: 98% entity attribution accuracy, 93% reduction in manual reporting effort, and production Agentic AI competitor-intelligence pipelines.

Core Skills

Cloud & Data Platforms: Azure (Synapse, ADF, Blob Storage, Key Vault, AKS, DevOps), Snowflake, AWS (Redshift), GCP (BigQuery), Databricks, Microsoft Fabric

Data Engineering: ETL/ELT Pipelines, DBT, SSIS, Delta Lake, Medallion Architecture, Data Modeling, Data Warehousing, Data Governance

Programming & Tools: Python, SQL, PySpark, Pandas, Async/Await, Selenium, Git, Docker, Kubernetes, N8N, pytest, GitHub Actions (CI/CD)

AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, Agentic AI, LLM Integration, Azure OpenAI GPT, LangChain, Claude Code, GitHub Copilot, Agent Skills

BI & Visualization: Power BI, SSRS, Paginated Reports, DAX, Excel (Power Query, VBA)

Databases: SQL Server, PostgreSQL, MongoDB, Oracle, Azure SQL, Snowflake

Professional Experience

Principal Advanced Analytics Engineer

ITWorx

Oct 2023 – Present
Cairo, Egypt (Hybrid)

- Spearheaded the ADX Digital Transformation for Abu Dhabi Securities Exchange (Core42), delivering data infrastructure modernisation for a leading UAE financial institution serving investors, regulators, and exchange operators.
- Developed a Python CLI tool that automated Power BI Paginated Report standardisation, cutting processing time from 4 hours to 15 minutes (93% reduction) across 190+ reports deployed to 4 environments.
- Optimised Azure SQL read queries for the Market Watch trading system, reducing query response times by ~30-50% and improving real-time data retrieval for exchange platform users.
- Restructured table schemas in Azure Synapse Dedicated SQL Pool, improving query performance and storage efficiency for the exchange's core data warehouse.
- Designed and implemented a Power BI report for real-time performance monitoring of the Operational Data Store Database (ODSDB), enabling proactive system management.
- Led presales activities and built proof-of-concept Agentic AI solutions using N8N, Azure OpenAI GPT, and LangChain, positioning the team for AI-driven automation opportunities.
- Serve as an instructor for AIWorx, ITWorx's company-wide AI enablement initiative, training 150+ developers to build with AI coding agents (Claude Code, GitHub Copilot) and Agentic AI workflows.
- Mentored 4-5 junior engineers on analytics engineering best practices, accelerating adoption of modern data engineering and ML techniques across the analytics function.

Tech: Azure Synapse Analytics, Azure SQL, Power BI, Paginated Reports, PostgreSQL, Python, PowerShell, N8N, Agentic AI

Senior Data Engineer

Raisa Energy LLC

Dec 2021 – Oct 2023
Cairo, Egypt (Hybrid)

- Led a team of 4 engineers in migrating the data warehouse from a multi-layer architecture to a proper Medallion architecture (Bronze/Silver/Gold) using Snowflake and DBT, improving data lineage and governance.
- Designed and built the Entity Correction Model in DBT/Snowflake, an automated system that corrects revenue and expense allocations to proper legal entities using exact and fuzzy matching — improving entity attribution accuracy from ~70% to ~98%.
- Architected and developed WEnergyDocsEL, a high-performance async Python ETL pipeline that extracts, downloads, and loads financial documents from the WEnergy portal to Azure Blob Storage and Snowflake — achieving 68x faster processing.

- Implemented security-first design with Azure Key Vault integration, custom exception hierarchy (15+ types), and retry logic with exponential backoff for production-grade reliability.

Tech: Snowflake, DBT, Python (async/await), Azure Blob Storage, Azure Key Vault, SQL Server

Business Insights Senior Lead

Jan 2019 – Dec 2021

— VOIS (Vodafone)

Smart Village, Giza

- Engineered a Python desktop application (PyQT6 + Selenium + Win32com API) to automate recurring operational processes, reducing manual intervention by 25% and aligning with stakeholder requirements for intuitive UX.
- Owned and maintained multiple Power BI dashboards for Pre-sales, Demand & Utilisation, and Work in Progress analysis, establishing data governance frameworks that ensured accuracy, integrity, and security.
- Partnered with cross-departmental teams to embed analytics into operational workflows, driving increased adoption of self-service dashboards across Pre-sales, Demand, and WIP functions.
- Built and maintained a scalable Oracle SQL Data Warehouse, ensuring reliable data management for enterprise reporting across multiple business units.
- Executed data migration from on-premises systems to AWS Redshift and Google BigQuery, enhancing scalability and query performance for growing analytical workloads.

Tech: Python, PyQt6, Selenium, Power BI, Oracle SQL, AWS Redshift, Google BigQuery, Win32com API

Senior Business Intelligence Developer

Dec 2017 – Jan 2019

Atos

Cairo, Egypt (Hybrid)

- Led technical pre-sales for Enterprise BI and data platform engagements, delivering solution demos and scoping proposals that drove new client acquisition and account expansion.
- Implemented MongoDB as Infrastructure-as-a-Service (IaaS) on Azure, deploying scalable and resilient databases for multiple environments and improving data storage reliability.
- Built an Elastic Cloud on Kubernetes (ECK) cluster on AKS for application resource monitoring, ensuring optimal resource utilisation across Azure services.
- Enhanced security by integrating Azure authentication mechanisms into custom Python packages, enabling secure access to hybrid on-premises and cloud resources.
- Served as primary analyst for the Microsoft Practice Team, delivering Power BI, Excel, and SQL Server reporting solutions that improved data visibility and operational decision-making across client engagements.

Tech: Azure, MongoDB, Kubernetes (AKS/ECK), Power BI, Python, SQL Server, Excel

Data Analyst

May 2015 – Dec 2017

Al Safy Group

New Cairo

- Built a multi-branch sales performance dashboard in Excel (Power Query, DAX, VBA) to track and compare KPIs across 60+ retail branches (Switch Plus, Mi Home, Orange stores), replacing manual reporting with an automated, self-refreshing model.
- Developed SQL queries against Microsoft Dynamics AX to extract sales transactions for KPI analysis, and automated commission calculations and cheque-issuing workflows using Excel Power Query and advanced formulas.

Tech: Excel (Power Query, DAX, VBA), SQL Server, Microsoft Dynamics AX

Key Projects

crm – Dynamics 365 CE Automation CLI (Open Source)

May 2026 – Present

- Authored and maintain an open-source Python CLI that operates Microsoft Dynamics 365 CE / Dataverse (on-premises NTLM and cloud OAuth) entirely from the shell, spanning 30 command groups across record CRUD, OData/FetchXML queries, metadata, solution lifecycle, forms, dashboards, security, and plug-in registration.
- Built a “customizations as code” workflow: declarative YAML desired-state with reviewable plan artifacts and dry-run previews, replacing manual GUI customisation with repeatable, auditable deployments.
- Engineered the CLI as an AI-agent surface (stable JSON envelopes, on-demand agent skill, published `11ms.txt`); 56K lines of source covered by 5,200+ automated tests, shipped across 160+ releases with a 175-page documentation site.

Tech: Python, Click, Dataverse Web API (OData v4), FetchXML, pytest, Docker, MkDocs Material, GitHub Actions

Links: crm-cli-docs.pages.dev | github.com/Gharib89/crm

Competitor AI Insight – Agentic AI Pipeline (ITWorx)

Jan 2025 – Present

- Designed and built a POC Agentic AI system that automatically discovers, scrapes, and analyses competitor intelligence across news, press releases, job postings, and RSS feeds from 28 sources — replacing manual monitoring with an AI-powered pipeline.
- Orchestrated 4 N8N workflows with Azure OpenAI GPT for structured extraction (13 fields per article), 6-topic classification with confidence scoring, sentiment analysis, and multi-engine scraping (HTTP, FlareSolvrr, Playwright, newspaper4k).
- Engineered a star-schema Azure SQL data warehouse and Docker Compose infrastructure (N8N, PostgreSQL, Playwright scraper, FlareSolvrr, Caddy) — currently tracking 10 competitors with zero-code extensibility.

Tech: N8N, Azure OpenAI GPT, Python, FastAPI, Playwright, Docker Compose, Azure SQL, PostgreSQL, LangChain

Entity Correction Model (Raisa Energy)

Feb 2023 – Sep 2023

- Built a multi-layer DBT model in Snowflake that resolves revenue and expense allocations against Jarvis ownership data, layering exact matching, normalised matching, and fuzzy matching so each record is settled at the cheapest tier that can decide it.
- Designed a 4-status matching flow (Matched / Fixed / Multiple / No Match) that resolves ~98.5% of records automatically and routes only ~1.5% to analyst review, making the remaining manual effort explicit and bounded rather than hidden.

Tech: Snowflake, DBT, SQL

WEnergyDocsEL – Document ETL Pipeline (Raisa Energy)

Mar 2022 – Nov 2022

- Engineered an async pipeline that scrapes document URIs from the WEnergy portal and moves invoices and cheques into Azure Blob Storage with metadata landed in Snowflake, sustaining 64 concurrent downloads and 12 concurrent uploads at 1,000+ documents/day (~3–5 minutes per 1,000).
- Implemented incremental and full-reload modes: incremental processes only documents changed since the last run and prunes orphans deleted upstream, keeping Azure and Snowflake a true mirror of the portal rather than an append-only copy.

Tech: Python (asyncio), Azure Blob Storage, Snowflake, Web Scraping

Certifications

- Microsoft Certified: Fabric Data Engineer Associate (DP-700)
- Microsoft Certified: Power BI Data Analyst Associate (PL-300)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- Azure Data Engineering Associate (DP-203) · Databricks Academy: SQL Analyst Associate · Google Cloud Data Engineer
- Udacity Nanodegrees: Data Engineering, Machine Learning Engineer, Cloud DevOps Engineer, Data Streaming, AI Programming with Python, Advanced Data Analysis
- Coursera Specializations: Deep Learning (DeepLearning.AI), Mathematics for Machine Learning

Education

Bachelor of Commerce

2008 – 2011

Ain Shams University, Cairo